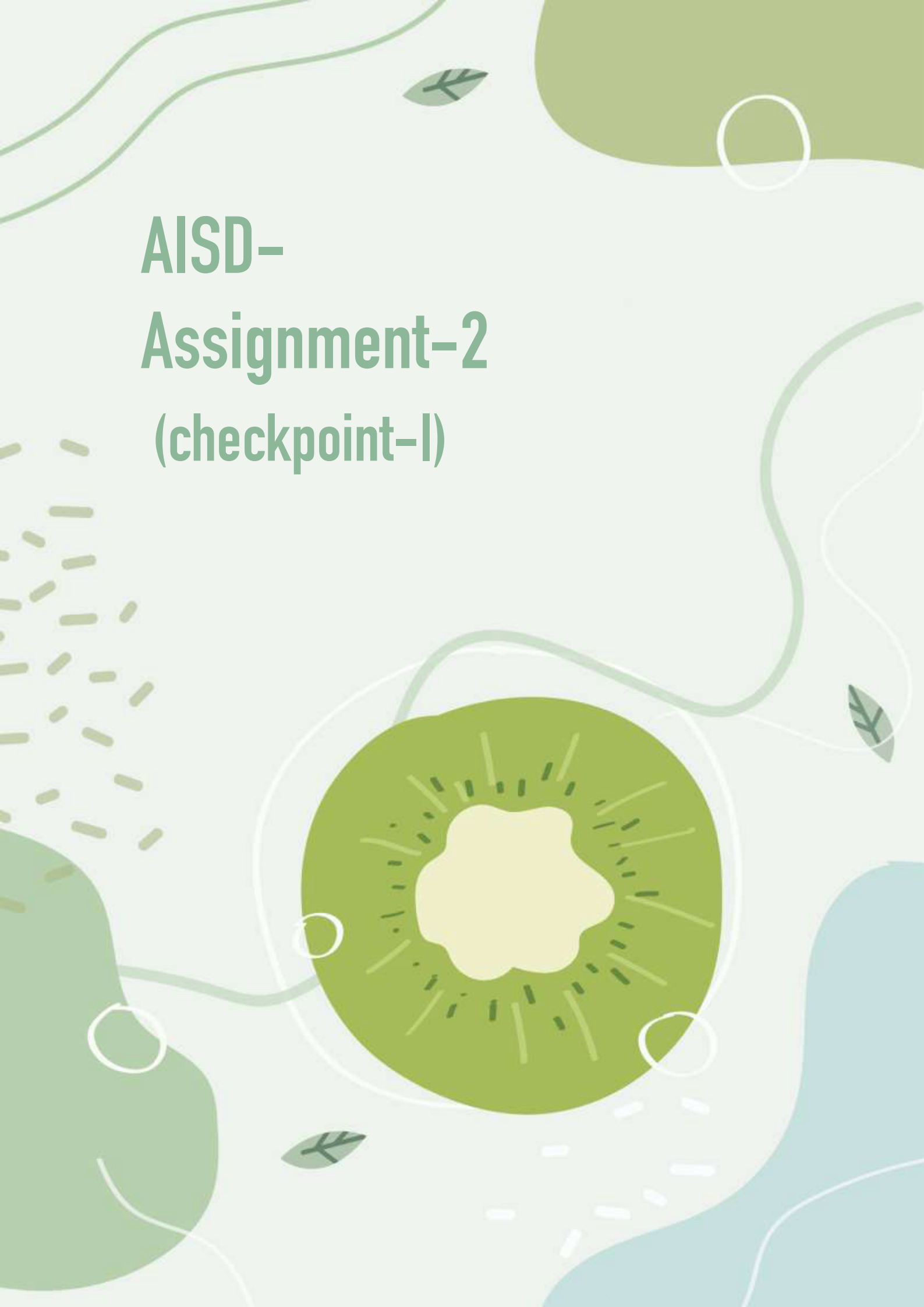
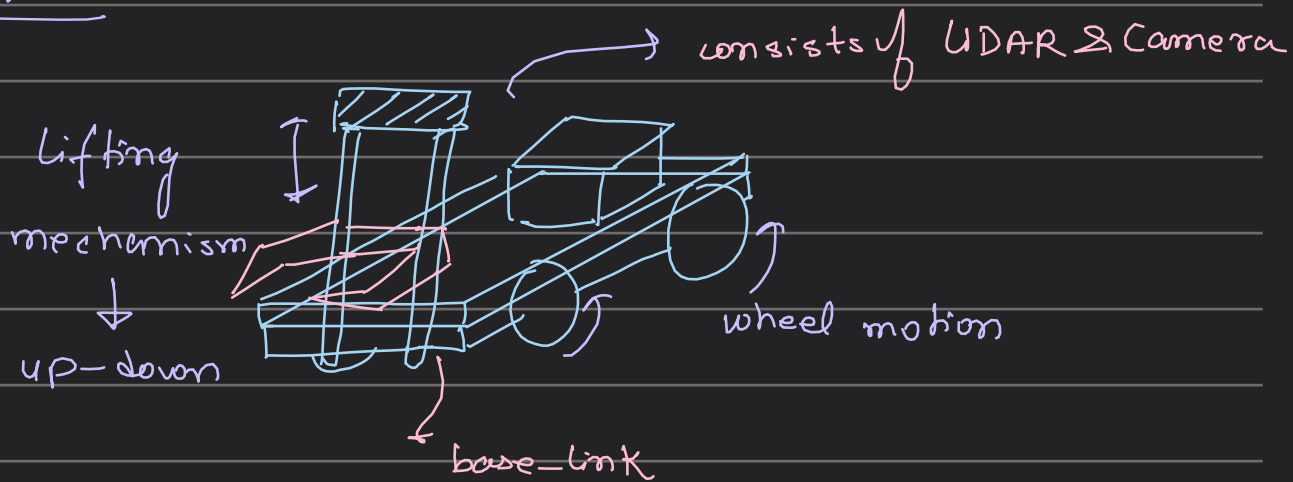


AISD- Assignment-2 (checkpoint-1)

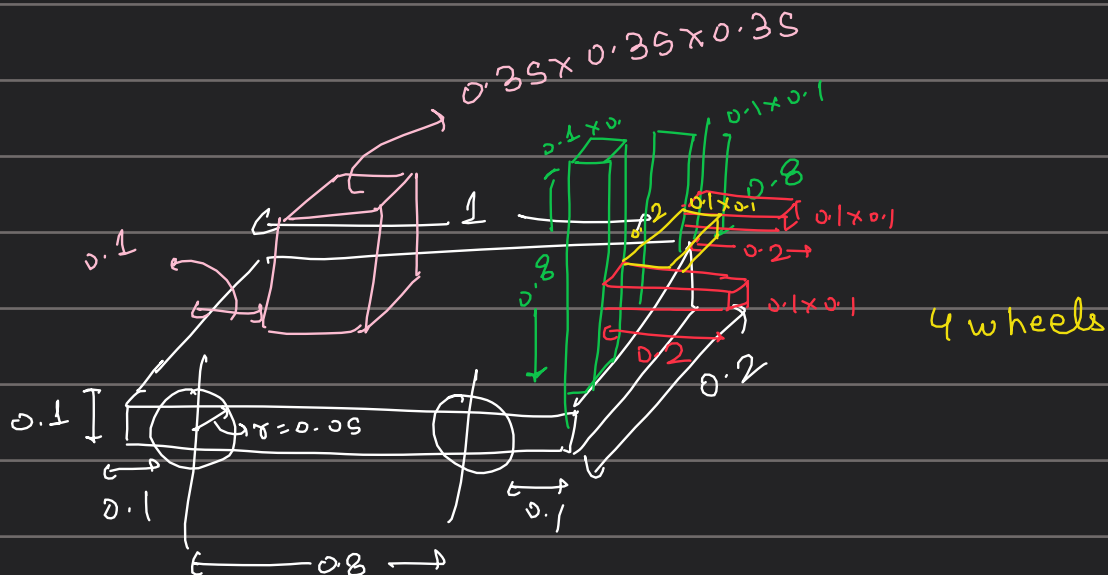
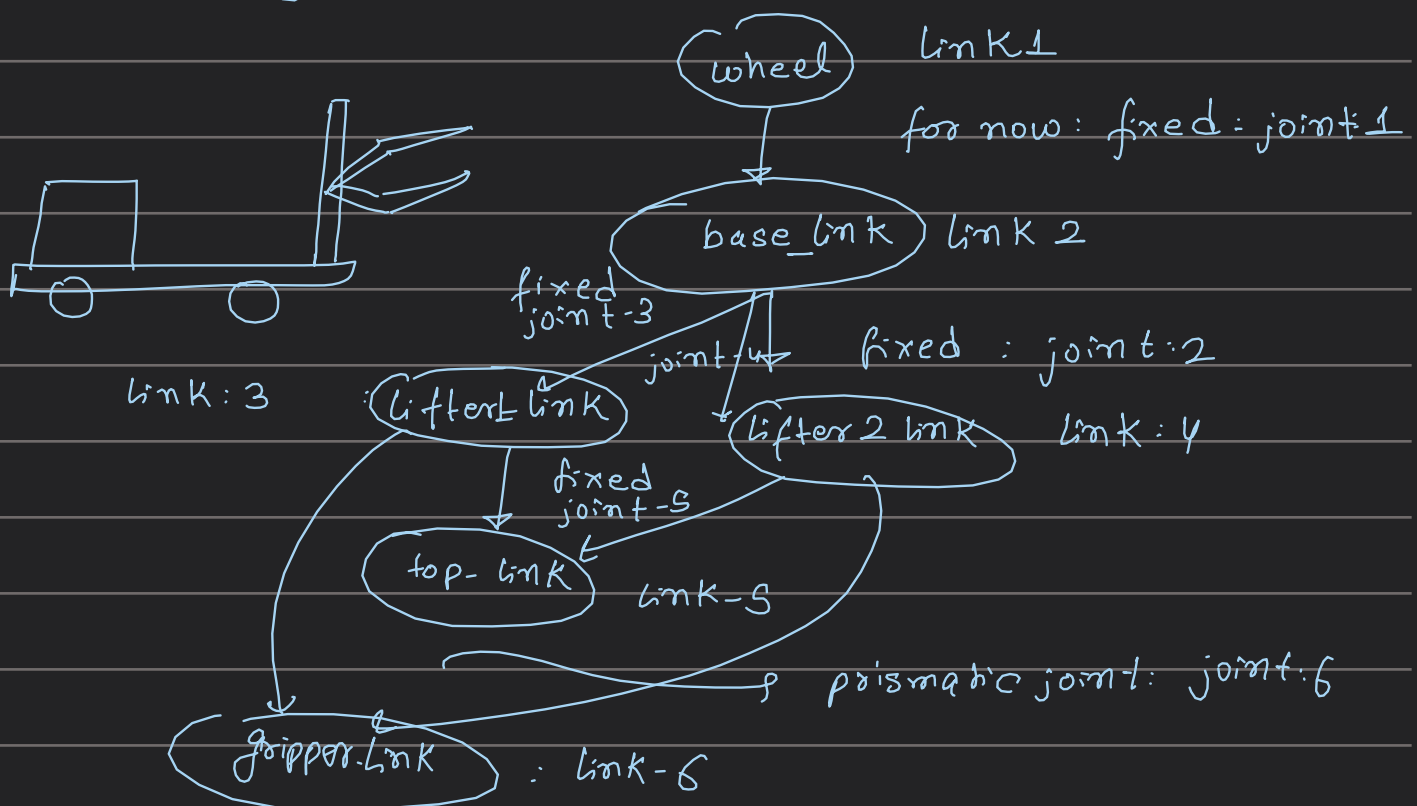


Assignment-2



Checkpoint-1: ① /joint_state

② /robot_state_publisher



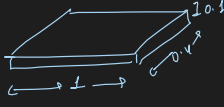
Rviz:

fixed_frame = "base_link"

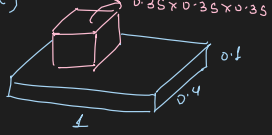
add: RobotModel $\rightarrow$ /robot_description

```
EXPLORER
  SRC
    > build
    > install
    > log
    > urdf_tutorials
      > launch
      > meshes
      > resource
      > rviz
      > test
      > urdf
        my_robot.urdf
      urdf_tutorials
        _init_.py
        package.xml
        setup.cfg
        setup.py
  my_robot.urdf
    urdf_tutorials
      > urdf
        my_robot.urdf
          1 <?xml version="1.0"?>
          2 <robot name = "my_robot">
          3 <link name = "base_link">
          4 <visual>
          5 <geometry>
          6 <box size = "1 0.4 0.1"/>
          7 </geometry>
          8 <material name="blue">
          9 <color rgba="0 0 0.8 1"/>
          10 </material>
          11 <origin xyz="0 0 0" rpy="0 0 0"/>
          12 <!--axis-->
          13 </visual>
          14 <collision>
          15 <origin xyz="0 0 0" rpy="0 0 0"/>
          16 <geometry>
          17 <box size = "1 0.4 0.1"/>
          18 </geometry>
          19 </collision>
          20 <inertial>
          21 <mass value = "10"/>
          22 <origin xyz="0 0 0" rpy="0 0 0"/>
          23 <inertia ixx="0.138" ixy="0" ixz="0" iyy="0.843" iyz="0" izz="0.967"/>
          24 </inertial>
          25 </link>
          26
          27 <link name = "weight_link">
          28 <visual>
          29 <geometry>
          30 <box size = "0.35 0.35 0.35"/>
          31 </geometry>
          32 <material name="red">
          33 <color rgba="0.8 0.8 0.1"/>
          34 </material>
          35 <origin xyz="0 0 0" rpy="0 0 0"/>
          36 <!--axis-->
          37 </visual>
          38 <collision>
          39 <origin xyz="0 0 0" rpy="0 0 0"/>
          40 <geometry>
          41 <box size = "0.5 0.5 0.5"/>
          42 </geometry>
          43 </collision>
```

Base link

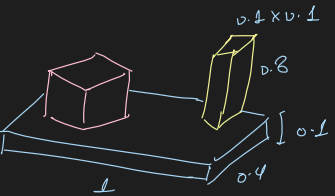


box (weight)



```
12 </geometry>
13 </collision>
14 </inertial>
15 <mass value = "5"/>
16 <origin xyz="0 0 0" rpy="0 0 0"/>
17 <inertia ixx="0.208" ixy="0" ixz="0" iyy="0.208" iyz="0" izz="0.208"/>
18 </inertial>
19 </link>
20
21 <joint name = "base_weight_joint" type = "fixed">
22 <parent link = "base_link"/>
23 <child link = "weight_link"/>
24 <origin xyz="-0.25 0 0.3" rpy="0 0 0"/>
25 <!--axis-->
26 </joint>
27
28 <link name = "lifter_1_link">
29 <visual>
30 <geometry>
31 <box size = "0.1 0.1 0.8"/>
32 </geometry>
33 <material name="yellow">
34 <color rgba="0.8 0.8 0.1"/>
35 </material>
36 <origin xyz="0 0 0" rpy="0 0 0"/>
37 <!--axis-->
38 </visual>
39 <collision>
40 <origin xyz="0 0 0" rpy="0 0 0"/>
41 <geometry>
42 <box size = "0.1 0.1 0.8"/>
43 </geometry>
44 </collision>
45 <inertial>
46 <mass value = "4"/>
47 <origin xyz="0 0 0" rpy="0 0 0"/>
48 <inertia ixx="0.003" ixy="0" ixz="0" iyy="0.215" iyz="0" izz="0.215"/>
49 </inertial>
```

farther link

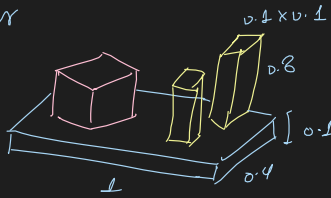


```

79 | </inertial>
80 | </link>
81 |
82 | <link name = "lifter_2_link">
83 |   <visual>
84 |     <geometry>
85 |       <box size = "0.1 0.1 0.8"/>
86 |     </geometry>
87 |     <material name="yellow">
88 |       <color rgba="0.8 0.8 0 1"/>
89 |     </material>
90 |     <origin xyz="0 0 0" rpy="0 0 0"/>
91 |     <!--axis-->
92 |   </visual>
93 |   <collision>
94 |     <origin xyz="0 0 0" rpy="0 0 0"/>
95 |     <geometry>
96 |       <box size = "0.1 0.1 0.8"/>
97 |     </geometry>
98 |   </collision>
99 |   <inertial>
100 |     <mass value = "4"/>
101 |     <origin xyz="0 0 0" rpy="0 0 0"/>
102 |     <inertia ixx="0.003" ixy="0" ixz="0" iyy="0.215" iyz="0" izz="0.215"/>
103 |   </inertial>
104 | </link>

```

↪ closer lifter



```

105 |
106 | <joint name = "base lifter 1 joint" type = "fixed">
107 |   <parent link = "base link"/>
108 |   <child link = "lifter_1 link"/>
109 |   <origin xyz="0.4 0.15 0.45" rpy="0 0 0"/>
110 |   <!--axis-->
111 | </joint>
112 |
113 | <joint name = "base lifter 2 joint" type = "fixed">
114 |   <parent link = "base link"/>
115 |   <child link = "lifter_2 link"/>
116 |   <origin xyz="0.4 -0.15 0.45" rpy="0 0 0"/>
117 |   <!--axis-->
118 | </joint>

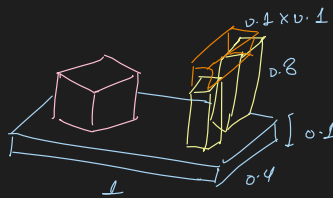
```

```

119 |
120 | <link name = "support link">
121 |   <visual>
122 |     <geometry>
123 |       <box size = "0.1 0.4 0.1"/>
124 |     </geometry>
125 |     <material name="orange">
126 |       <color rgba="1 0.5 0 1"/>
127 |     </material>
128 |     <origin xyz="0 0 0" rpy="0 0 0"/>
129 |     <!--axis-->
130 |   </visual>
131 |   <collision>
132 |     <origin xyz="0 0 0" rpy="0 0 0"/>
133 |     <geometry>
134 |       <box size = "0.4 0.1 0.1"/>
135 |     </geometry>
136 |   </collision>
137 |   <inertial>
138 |     <mass value = "2"/>
139 |     <origin xyz="0 0 0" rpy="0 0 0"/>
140 |     <inertia ixx="0.002" ixy="0" ixz="0" iyy="0.014" iyz="0" izz="0.014"/>
141 |   </inertial>
142 | </link>

```

↪ top (cap)



```

143 |
144 | <joint name = "lifter_1 support joint" type = "fixed">
145 |   <parent link = "lifter_1 link"/>
146 |   <child link = "support link"/>
147 |   <origin xyz="0 -0.15 0.45" rpy="0 0 0"/>
148 |   <!--axis-->
149 | </joint>

```

```

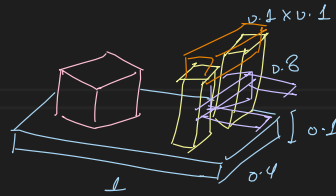
150 |
151 | <link name="gripper_link">
152 |   <!-- left fork -- long prong sticking forward -->
153 |   <visual>
154 |     <origin xyz="0.2 0.1 0"/>
155 |     <geometry><box size="0.35 0.05 0.04"/></geometry>
156 |     <material name="light_pink"><color rgba="1 0.75 0.8 1"/></material>
157 |   </visual>

```

```

58 <!-- right fork -- long prong sticking forward -->
59 <visual>
60   <origin xyz="0.2 -0.1 0"/>
61   <geometry><box size="0.35 0.05 0.04"/></geometry>
62   <material name="light_pink"><color rgba="1 0.75 0.8 1"/></material>
63 </visual>
64 <!-- back plate connecting both forks -->
65 <visual>
66   <origin xyz="0 0 0"/>
67   <geometry><box size="0.05 0.3 0.20"/></geometry>
68   <material name="light_pink"><color rgba="1 0.75 0.8 1"/></material>
69 </visual>
70
71 <collision>
72   <origin xyz="0.3 0.1 0"/>
73   <geometry><box size="0.6 0.06 0.04"/></geometry>
74 </collision>
75 <collision>
76   <origin xyz="0.3 -0.1 0"/>
77   <geometry><box size="0.6 0.06 0.04"/></geometry>
78 </collision>
79 <collision>
80   <origin xyz="0 0 0"/>
81   <geometry><box size="0.06 0.3 0.25"/></geometry>
82 </collision>
83
84 <inertial>
85   <mass value="1"/>
86   <origin xyz="0 0 0" rpy="0 0 0"/>
87   <inertia ixx="0.002" ixy="0" ixz="0" iyy="0.001" iyz="0" izz="0.001"/>
88 </inertial>
89 </link>

```



```

1 <joint name="lifter gripper joint" type="prismatic">
2   <parent link="support link"/>
3   <child link="grripper link"/>
4   <origin xyz="0.05 0 -0.1" rpy="0 0 0"/>
5   <axis xyz="0 0 -1"/>
6   <limit lower="0" upper="0.3" effort="10" velocity="0.5"/>
7 </joint>
8
9 <link name = "wheel 1 link">
10  <visual>
11    <geometry>
12      <cylinder length="0.04" radius="0.08"/>
13    </geometry>
14    <material name="black">
15      <color rgba="0 0 0 1"/>
16    </material>
17    <origin xyz="0 0 0" rpy="1.5708 0 0"/>
18    <!--axis-->
19  </visual>
20  <collision>
21    <origin xyz="0 0 0" rpy="1.5708 0 0"/>
22    <geometry>
23      <cylinder length="0.04" radius="0.08"/>
24    </geometry>
25  </collision>
26  <inertial>
27    <mass value = "1"/>
28    <origin xyz="0 0 0" rpy="0 0 0"/>
29    <inertia ixx="0.0006" ixy="0" ixz="0" iyy="0.0006" iyz="0" izz="0.00125"/>
30  </inertial>
31 </link>
32
33 <link name = "wheel 2 link">
34  <visual>
35    <geometry>
36      <cylinder length="0.04" radius="0.08"/>
37    </geometry>
38    <material name="black">
39      <color rgba="0 0 0 1"/>
40    </material>
41    <origin xyz="0 0 0" rpy="1.5708 0 0"/>
42    <!--axis-->
43  </visual>

```



```

</visual>
<collision>
  <origin xyz="0 0 0" rpy="1.5708 0 0"/>
  <geometry>
    <cylinder length="0.04" radius="0.08"/>
  </geometry>
</collision>
<inertial>
  <mass value = "1"/>
  <origin xyz="0 0 0" rpy="0 0 0"/>
  <inertia ixx="0.0006" ixy="0" ixz="0" iyy="0.0006" izy="0" izz="0.00125"/>
</inertial>
</link>

```

```

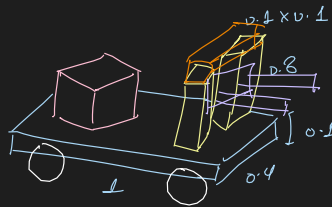
<link name = "wheel_3_link">
  <visual>
    <geometry>
      <cylinder length="0.04" radius="0.08"/>
    </geometry>
    <material name="black">
      <color rgba="0 0 0 1"/>
    </material>
    <origin xyz="0 0 0" rpy="1.5708 0 0"/>
    <!--axis-->
  </visual>
  <collision>
    <origin xyz="0 0 0" rpy="1.5708 0 0"/>
    <geometry>
      <cylinder length="0.04" radius="0.08"/>
    </geometry>
  </collision>
  <inertial>
    <mass value = "1"/>
    <origin xyz="0 0 0" rpy="0 0 0"/>
    <inertia ixx="0.0006" ixy="0" ixz="0" iyy="0.0006" izy="0" izz="0.00125"/>
  </inertial>
</link>

```

```

171 <link name = "wheel_4 link">
172 <visual>
173   <geometry>
174     <cylinder length="0.04" radius="0.08"/>
175   </geometry>
176   <material name="black">
177     <color rgba="0 0 0 1"/>
178   </material>
179   <origin xyz="0 0 0" rpy="1.5708 0 0"/>
180   <!--axis-->
181 </visual>
182 <collision>
183   <origin xyz="0 0 0" rpy="1.5708 0 0"/>
184   <geometry>
185     <cylinder length="0.04" radius="0.08"/>
186   </geometry>
187 </collision>
188 <inertial>
189   <mass value = "1"/>
190   <origin xyz="0 0 0" rpy="0 0 0"/>
191   <inertia ixx="0.0006" ixy="0" ixz="0" iyy="0.0006" izy="0" izz="0.00125"/>
192 </inertial>
193 </link>

```



```

194 <joint name = "base wheel 1 joint" type = "fixed">
195   <parent link = "base_link"/>
196   <child link = "wheel_1 link"/>
197   <origin xyz="0.4 0.2 -0.1" rpy="0 0 0"/>
198   <!--axis-->
199 </joint>

```

```

202 <joint name = "base wheel 2 joint" type = "fixed">
203   <parent link = "base_link"/>
204   <child link = "wheel_2 link"/>
205   <origin xyz="0.4 -0.2 -0.1" rpy="0 0 0"/>
206   <!--axis-->
207 </joint>

```

```

307 </joint>
308 <joint name = "base wheel 3 joint" type = "fixed">
309   <parent link = "base_link"/>
310   <child link = "wheel_3 link"/>
311   <origin xyz="0.4 0.2 -0.1" rpy="0 0 0"/>
312   <!--axis-->
313 </joint>

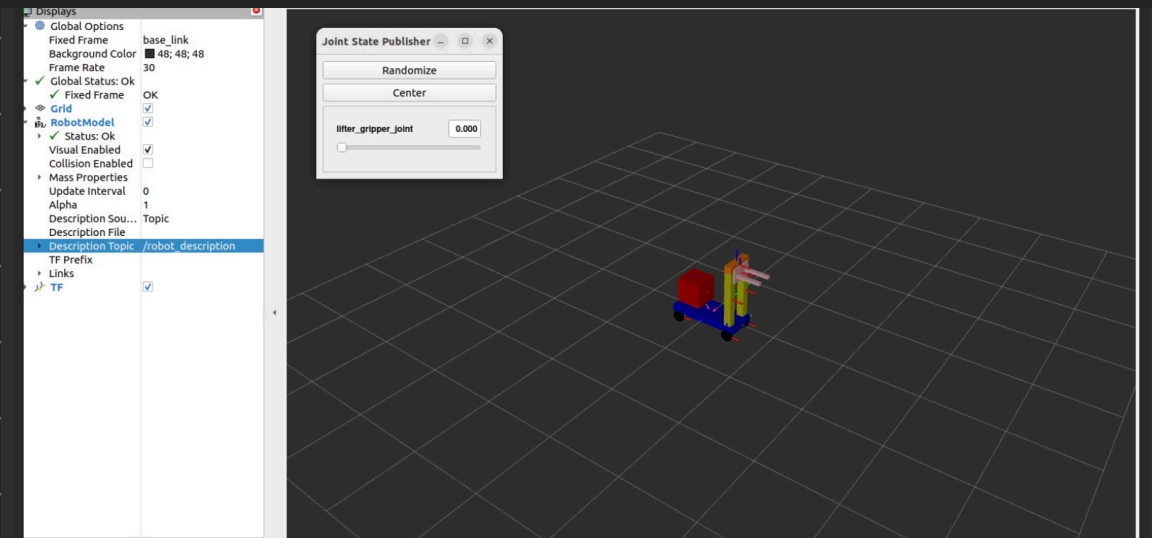
```

```

315 <joint name = "base wheel 4 joint" type = "fixed">
316   <parent link = "base_link"/>
317   <child link = "wheel_4 link"/>
318   <origin xyz="0.4 -0.2 -0.1" rpy="0 0 0"/>
319 </joint>
320 </robot>
321
322
323

```

Rviz2



terminal-1:

Command:

```
ros2 run robot_state_publisher robot_state_publisher --ros-args  
-p robot_description:="$(cat ~/.AISD_ws/src/urdf-tutorials/urdf/  
my-robot.urdf)"
```

terminal-2:

Command:

```
ros2 run joint_state_publisher_gui joint_state_publisher_gui
```

terminal-3:

Command: rviz2

terminal-4:

Command: ros2 topic list

to see the functionality

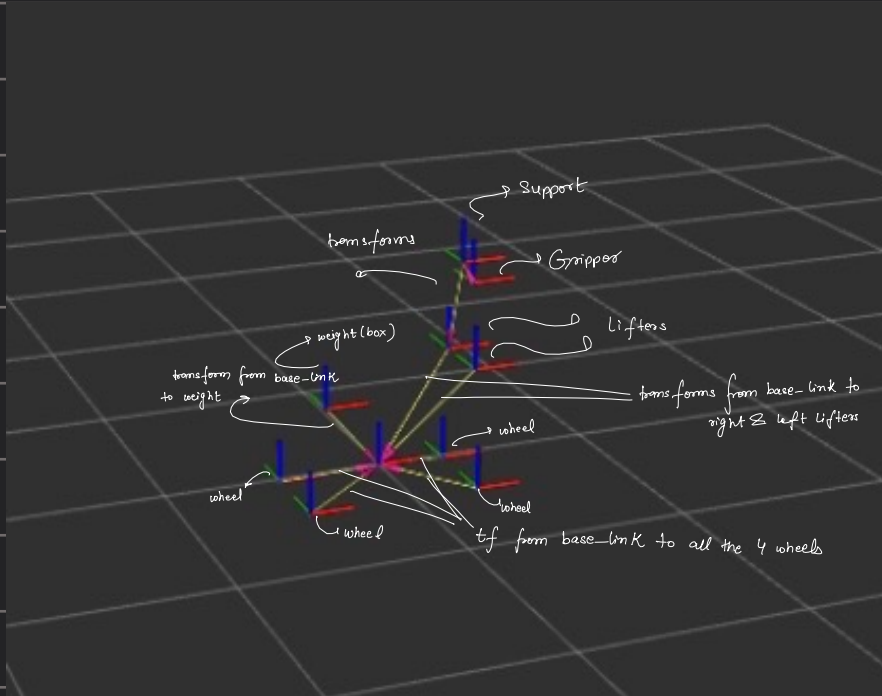
Reviz 2: ① Fixed Frame = "base-link"

② Add → By Display Type → Robot Model

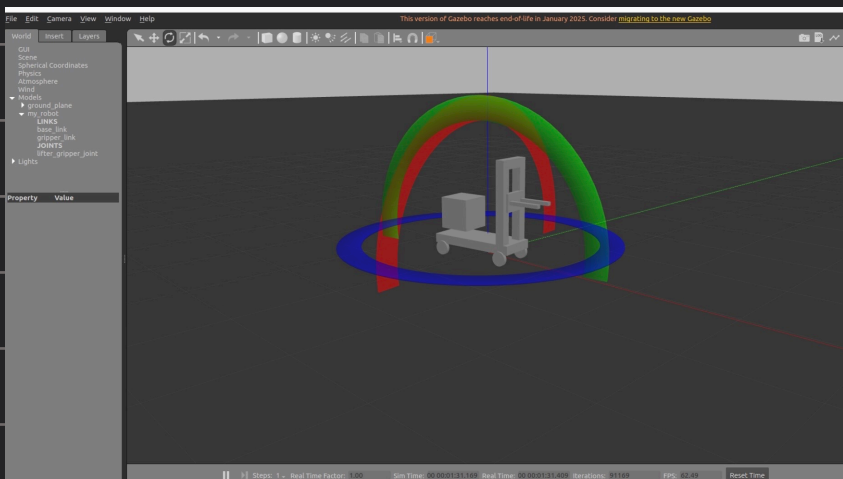
↓
choose topic as /robot-description

③ Add → By Display Type → TF

↳ all the transforms will become visible



Gazebo



terminal-1:

Command:

```
ros2 run robot_state_publisher robot_state_publisher --ros-args  
-p robot_description:="$(cat ~/AISD-ws/src/urdf-tutorials/urdf/  
my-robot.urdf)"
```


terminal-2:

Command: `ros2 launch gazebo-ros gazebo.launch.py`

terminal-3:

Command: `ros2 run gazebo-ros spawn-entity.py -topic
robot-description -entity my-robot`